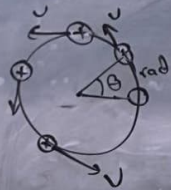


بسم الله الرحمن الرحيم

ex. 2.6

An electron accelerated from rest by 350 V in $B = 0.2 \text{ T}$.

$$KE = qV = 1.6 \times 10^{-19} \times 350 = 560 \times 10^{-19} \text{ J}$$



(a) Linear velocity:
$$v = \sqrt{\frac{2qV}{m}} = \sqrt{\frac{2 \times 560 \times 10^{-19}}{9.11 \times 10^{-31}}} = 1.1 \times 10^7 \text{ m/s}$$

(b) Angular Velocity:
$$\omega = \frac{qB}{m} = \frac{1.6 \times 10^{-19} \times 0.2}{9.11 \times 10^{-31}} = 3.5 \times 10^{10} \text{ rad/s}$$

(c) radius:
$$r = \frac{v}{\omega} = \frac{1.1 \times 10^7}{3.5 \times 10^{10}} = 3 \times 10^{-4} \text{ m}$$

$$r = \frac{mv}{qB}$$